

TrES-3b

R-0423

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250808 · created 2026-07-19T09:03:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460895.7906 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.055
Transit depth (Rp/Rs)^2	2.62 +/- 1.78 [%]
Orbital Inclination (inc)	79.6 +/- 2.86
Airmass coefficient 1 (a1)	0.955 +/- 0.063
Airmass coefficient 2 (a2)	0.037 +/- 0.083
Scatter in the residuals of the lightcurve fit is	6.53 %
Best Comparison Star	#2 - [467, 472]
Optimal Aperture	5.55
Optimal Annulus	8.79
Transit Duration (day)	0.027 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.055 vs 0.16309 (deviation 0.02σ)
Duration fit vs published	0.027 d vs 0.0567 d (deviation 1.0σ)
Mid-time O–C	-0.1 min
Depth SNR	2.9
Residual scatter	6.53 %

Light curve

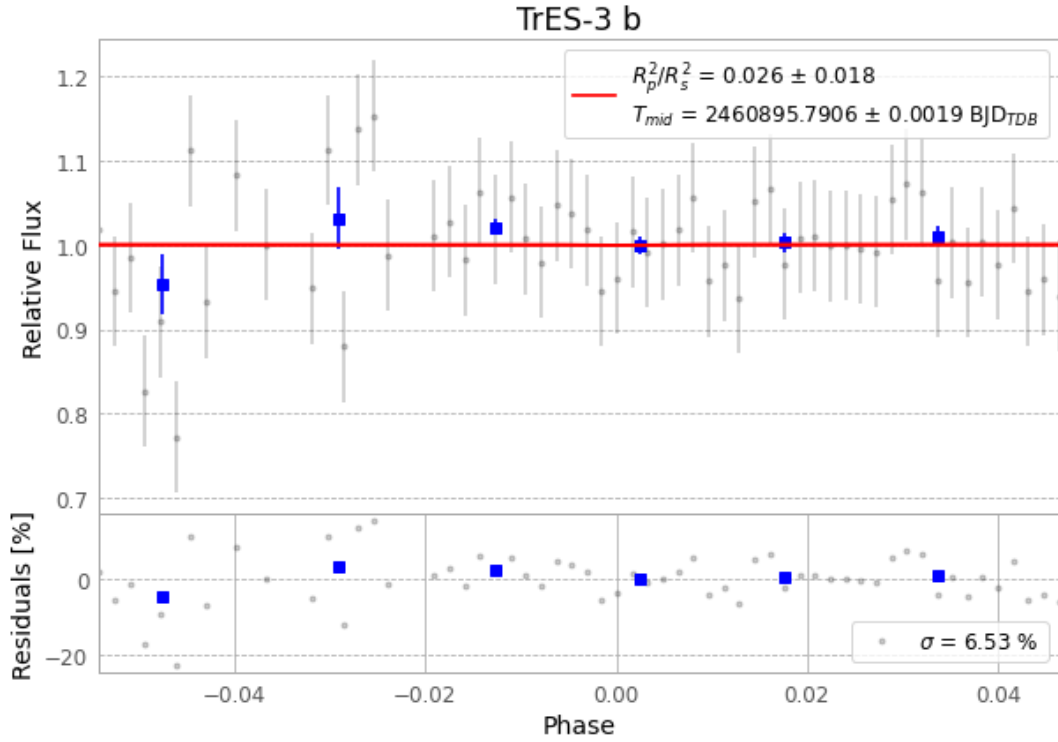


Figure 1 — FinalLightCurve_TrES-3 b_2025-08-07.png (SHA-256 dfde4576730c2bb4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b123f1a4411353af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), TRES-3250808051215.FITS → TRES-3250808082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250808.log (4d162adc5c95...), FinalLightCurve_TrES-3 b_2025-08-07.png (dfde4576730c...), FinalLightCurve_TrES-3 b_2025-08-07.csv (7e310e13a84b...)

Compute resources

Wall time 260.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 09:03Z.